

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0889

Mt Poso 138 kV
April 28, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0889. The Interconnection Customer proposes a 178.1 MW wind facility interconnecting at the Mt Poso 138 kV substation in the New Jersey area. The request was received on November 05, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 328 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$49,920,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The West Orange – Magnolia 1708 138 kV line is overloaded to 325.6% of its 238 MVA emergency rating for the loss of the Sne - Sansum N Elec – Multitrade Of Pittsylvania Lp 138 kV. The project contributes 46.3% DFAX (82.5 MW) to this overload.

Magnolia 1708 – White Rock-Nike 138 kV Line: 319.7% loading (243 MVA) following the outage of the Crawfordsville Power Plant – Pine Tree Acres 138 kV line; DFAX = 22.6%.

The Dam Heights – Babine Lake 138 kV is overloaded to 277.2% of its 226 MVA emergency rating for the loss of the Magnolia 1708 – West Orange Substation 138 kV. The project contributes 10.6% DFAX (18.9 MW) to this overload.

Jan Phyl – Babine Lake Substation 138 kV: 267.5% loading (236 MVA) following the outage of the Magnolia 1708 – W Orange 138 kV; DFAX = 15.0%.

For the loss of W Orange Substation – Magnolia 1708 138 kV, the Jan Phyl – Sne - Sansum N Elec 138 kV line loads to 240.3% of its 288 MVA rating. Project DFAX: 15.0%.

For the loss of Magnolia 1708 – W Orange Substation 138 kV line, the Sne - Sansum N Elec – Multitrade Of Pittsylvania Lp Substation 138 kV loads to 219.0% of its 332 MVA rating. Project DFAX: 15.0%.

White Rock-Nike – 15T St 138 kV: 172.2% loading (253 MVA) following the outage of the White Rock-Nike – Hudson 138 kV; DFAX = 22.9%.

Mt Poso – West Orange 138 kV: 171.2% loading (226 MVA) following the outage of the Hilltop Alte – Blancard 138 kV line; DFAX = 57.4%.

The Locust Grove – Shredder 138 kV is overloaded to 168.7% of its 289 MVA emergency rating for the loss of the White Rock-Nike – Magnolia 1708 138 kV. The project contributes 19.2% DFAX (34.3 MW) to this overload.

For the loss of White Rock-Nike – Magnolia 1708 138 kV Line, the Shredder – Roxboro 138 kV L loads to 151.3% of its 243 MVA rating. Project DFAX: 12.4%.

For the loss of Hudson – White Rock-Nike 138 kV, the Cla St – 15Th Street 138 kV loads to 146.3% of its 275 MVA rating. Project DFAX: 22.9%.

For the loss of 15Th Street – White Rock-Nike Substation 138 kV, the White Rock-Nike – Hudson 138 kV loads to 141.9% of its 244 MVA rating. Project DFAX: 14.5%.

For the loss of W Orange – Magnolia 1708 138 kV line, the Dam Hts – Artesian 138 kV Line loads to 139.0% of its 228 MVA rating. Project DFAX: 6.1%.

The Brown – Mjmeuc Geerating Sta 1 138 kV line is overloaded to 137.2% of its 254 MVA emergency rating for the loss of the West Orange – Magnolia 1708 138 kV. The project contributes 39.6% DFAX (70.6 MW) to this overload.

The Multitrade Of Pittsylvania Lp – Northeastern 138 kV is overloaded to 135.6% of its 300 MVA emergency rating for the loss of the W Orange – Magnolia 1708 138 kV. The project contributes 12.6% DFAX (22.4 MW) to this overload.

The Adams St – Hudson 138 kV Line is overloaded to 134.1% of its 202 MVA emergency rating for the loss of the 15Th Street – White Rock-Nike 138 kV. The project contributes 14.5% DFAX (25.8 MW) to this overload.

The Orange – Suntree 138 kV is overloaded to 127.8% of its 251 MVA emergency rating for the loss of the Magnolia 1708 – West Orange 138 kV line. The project contributes 16.4% DFAX (29.2 MW) to this overload.

Crawfordsville Power Plant – Roxboro Substation 138 kV: 126.0% loading (257 MVA) following the outage of the Magnolia 1708 Substation – White Rock-Nike 138 kV; DFAX = 12.4%.

For the loss of Magnolia 1708 Substation – W Orange 138 kV, the Brown – Aeropuerto 138 kV loads to 124.4% of its 323 MVA rating. Project DFAX: 39.6%.

For the loss of Magnolia 1708 – W Orange 138 kV, the Suntree Substation – W Caldwell 138 kV loads to 124.1% of its 243 MVA rating. Project DFAX: 16.4%.

The Marion Drive – Mjmeuc Generating Station 1 138 kV L is overloaded to 123.2% of its 262 MVA emergency rating for the loss of the Magnolia 1708 – West Orange 138 kV line. The project contributes 39.6% DFAX (70.6 MW) to this overload.

Aeropuerto – Hilltop Alte 138 kV: 121.9% loading (348 MVA) following the outage of the Magnolia 1708 Substation – W Orange 138 kV; DFAX = 39.6%.

For the loss of Dam Hts – Athenia 138 kV, the Wasco – Artesian 138 kV loads to 119.0% of its 228 MVA rating. Project DFAX: 5.2%.

The West Caldwell – Northeastern 138 kV L is overloaded to 116.9% of its 229 MVA emergency rating for the loss of the Magnolia 1708 – W Orange 138 kV. The project contributes 16.4% DFAX (29.2 MW) to this overload.

The 5075 Hosp. Damas Substation – Adams St 138 kV is overloaded to 113.2% of its 219 MVA emergency rating for the loss of the White Rock-Nike – 15Th St Substation 138 kV. The project contributes 14.5% DFAX (25.8 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Magnolia 1708 – West Orange Substation 138 kV	Rebuild line with higher-capacity conductor	\$802,000	100% (\$802,000)
Magnolia 1708 – White Rock-Nike 138 kV Line	Rebuild line with higher-capacity conductor	\$1,478,000	100% (\$1,478,000)
Dam Heights – Babine Lake 138 kV	Rebuild line with higher-capacity conductor	\$644,000	100% (\$644,000)
Jan Phyl – Babine Lake 138 kV	Rebuild line with higher-capacity conductor	\$1,844,000	100% (\$1,844,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jan Phyl – Sne - Sansum N Elec 138 kV line	Rebuild line with higher-capacity conductor	\$1,523,000	100% (\$1,523,000)
Sne - Sansum N Elec – Multitrade Of Pittsylvania Lp Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,309,000	100% (\$1,309,000)
White Rock-Nike – 15T St 138 kV	Rebuild line with higher-capacity conductor	\$1,485,000	100% (\$1,485,000)
Mt Poso – West Orange 138 kV	Rebuild line with higher-capacity conductor	\$5,575,000	100% (\$5,575,000)
Locust Grove – Shredder 138 kV	Rebuild line with higher-capacity conductor	\$1,653,000	100% (\$1,653,000)
Roxboro – Shredder 138 kV	Rebuild line with higher-capacity conductor	\$3,352,000	100% (\$3,352,000)
Clay St Substation – 15Th St 138 kV Line	Rebuild line with higher-capacity conductor	\$2,226,000	100% (\$2,226,000)
Hudson Substation – White Rock-Nike Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,217,000	100% (\$1,217,000)
Artesian – Dam Hts Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,222,000	100% (\$1,222,000)
Brown – Mjmeuc Geerating Sta 1 138 kV line	Rebuild line with higher-capacity conductor	\$2,487,000	100% (\$2,487,000)
Northeastern – Multitrade Of Pittsylvania Lp 138 kV	Rebuild line with higher-capacity conductor	\$3,689,000	100% (\$3,689,000)
Adams St – Hudson 138 kV	Rebuild line with higher-capacity conductor	\$1,376,000	100% (\$1,376,000)
Orange – Suntree 138 kV L	Rebuild line with higher-capacity conductor	\$1,477,000	100% (\$1,477,000)
Crawfordsville Power Plant – Roxboro Substation 138 kV	Rebuild line with higher-capacity conductor	\$1,777,000	100% (\$1,777,000)
Brown – Aeropuerto 138 kV	Rebuild line with higher-capacity conductor	\$2,047,000	100% (\$2,047,000)
Suntree Substation – W Caldwell 138 kV	Rebuild line with higher-capacity conductor	\$3,081,000	100% (\$3,081,000)
Marion Drive – Mjmeuc Generating Sta 1 138 kV line	Rebuild line with higher-capacity conductor	\$1,976,000	100% (\$1,976,000)
Hilltop Alte – Aeropuerto 138 kV	Rebuild line with higher-capacity conductor	\$1,503,000	100% (\$1,503,000)
Artesian – Wasco 138 kV	Rebuild line with higher-capacity conductor	\$1,310,000	100% (\$1,310,000)
West Caldwell – Northeastern 138 kV L	Rebuild line with higher-capacity conductor	\$4,192,000	100% (\$4,192,000)
5075 Hosp. Damas Substation – Adams St 138 kV	Rebuild line with higher-capacity conductor	\$675,000	100% (\$675,000)
	Total allocated to project		\$49,920,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.